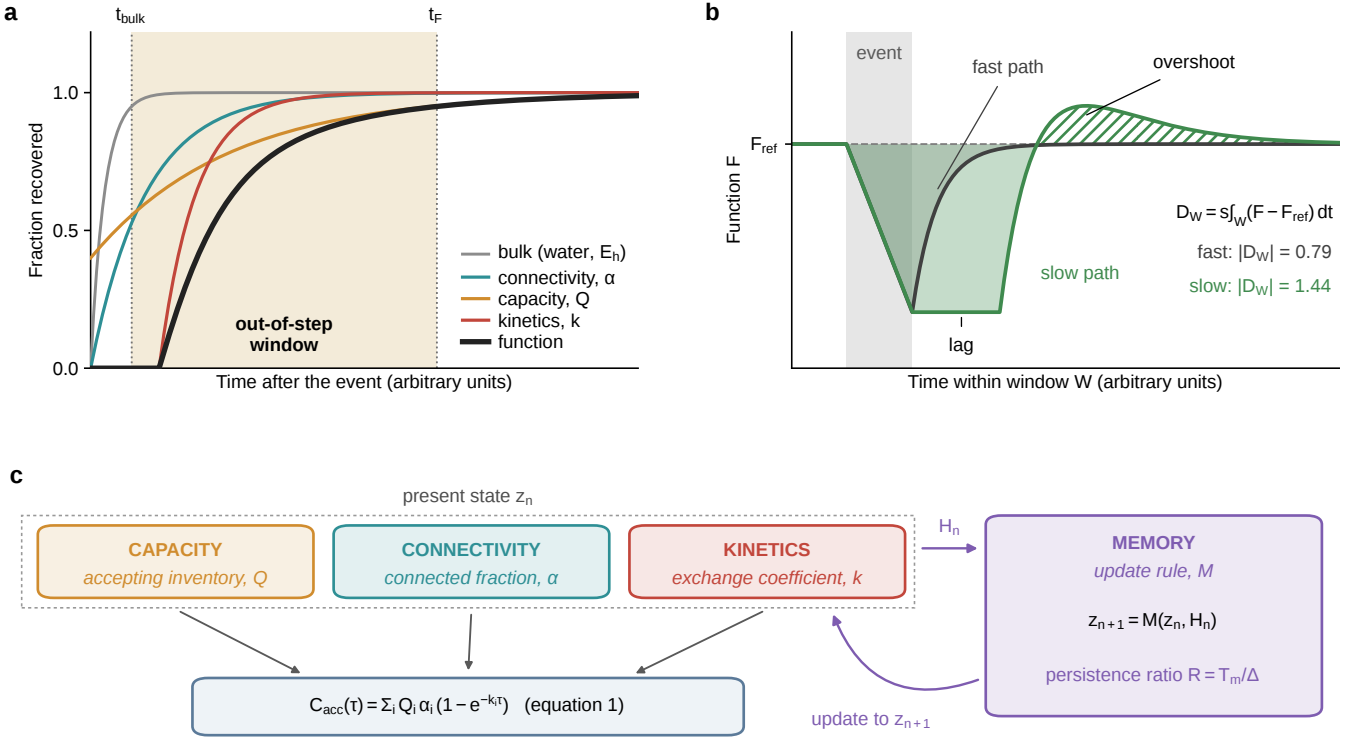

a
$t_{bulk}$
$t_F$
1.0
0.5
0.0
Fraction recovered
out-of-step window
bulk (water, $E_h$)
connectivity, α
capacity, Q
kinetics, k
function
Time after the event (arbitrary units)
b
event
fast path
overshoot
$F_{ref}$
Function F
$D_W = s\int_W (F - F_{ref})\,dt$
fast: $|D_W|$ = 0.79
slow: $|D_W|$ = 1.44
slow path
lag
Time within window W (arbitrary units)
c
present state $z_n$
CAPACITY
accepting inventory, Q
CONNECTIVITY
connected fraction, α
KINETICS
exchange coefficient, k
$H_n$
MEMORY
update rule, M
$z_{n+1} = M(z_n, H_n)$
persistence ratio $R = T_m/\Delta$
$C_{acc}(\tau) = \Sigma_i Q_i \alpha_i (1 - e^{-k_i \tau})$ (equation 1)
update to $z_{n+1}$